CYBER INTRUSION ANALYSIS, DETECTION AND CLASSIFICATION

[DERIVED FROM A SCHOOL PROJECT]

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About me:

The developer behind this project is a Bachelor of Computer Science (Hons) in Artificial Intelligence undergraduate student at Asia Pacific University of Technology and Innovation. He is also a certified Artificial Intelligence Engineer with certificates issued by Rocheston and Asia Pacific University of Technology of Technology and Innovation.

Choolwe Bright is passionate about Artificial Intelligence and Machine Learning. AI and ML are essential fields of study pivoted towards the future direction of data science and data analytics. Recent developments have demonstrated the automation of time-consuming processes such as data cleaning and data preparation of massive datasets and big data. Moreover, these technologies have enhanced predictive modelling, improved accuracy and strengthened feature engineering and has amplified our capacity to gain meaningful insights from data.

With the knowledge, skills and expertise acquired through his studies and professional training, the developer directed this project towards analysing data and showcasing the different ways machine learning can be utilized in data analysis at a small scale. This project reflects a practical application of machine learning concepts and an eagerness to explore the potential of AI-driven solutions in modern data analysis.

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1.0 Introduction

The increasing reliance on digital networks has significantly expanded the attack surface for cyber-threats, making intrusion detection systems is essential for maintaining network security. Modern intrusion detection approaches rely heavily on analysing network patterns and identifying patterns that may indicate malicious activity.

This project analyses the UNSW-NB15 dataset, a contemporary benchmark dataset designed for evaluating intrusion detection techniques. The dataset contains detailed network flow records representing both benign traffic and various types of cyber-attacks.

The primary objective of this project is to explore the dataset to identify patterns and relationships between network features such as service and dload. To achieve this the project undergoes a structured pipeline of data preprocessing and cleaning, exploratory data analysis, statistical hypothesis testing and machine learning modelling to evaluate the association of different variables to the presence of cyber-attacks.

Feature engineering techniques such as log transformations and anomaly flags were introduced to capture abnormal traffic behaviour, mostly pertaining to low-rate attacks and irregular download activity. Statistical analysis and predictive modelling, including logistic regression and random forest classification, were then used to evaluate the significance and predictive strength of these variables.

Through this approach, the project aims to identify meaningful traffic indicators that can support the design of more effective intrusion detection mechanisms.

2.0 Data Description:

This analysis is based on the UNSW-NB15 dataset which is one of the popular modern intrusion detection benchmarks created by the IXIA PerfectStorm traffic generator. It has a combination of legitimate (benign) network traffic and nine distinct types of cyber-attacks.

1. dload (Destination Load) or (Download Rate)
- Numeric
- Measures the rate of transferred data to the destination.
2. service
- Categorical
- Network service used (e.g. HTTP, FTP, DNS).

Dependant variable - Label:

- Categorical (0 = Normal, 1 = Attack)
- Is used to show whether a flow is malicious or benign.

Summary of Relevance:

The variables chosen are behaviour and traffic indicators that are very helpful in differentiating:

- normal vs. attack traffic.
- types of attack patterns.

2.0 Hypothesis and Objectives

2.1 Hypothesis:

There is a significant impact towards the presence of cyber-attacks from the combination service and dload..

2.2 Objectives:

1. To determine whether service type and download rates (dload and other variants) significantly influence the presence of cyber-attacks.

3.1 Data Import:

Firstly, the UNSW-NB15 data set is read using the read.csv () function, the variable **df** is initialized to initialize the data frame.

```
#data import
fileUrl<-"C:/Users/HP 15/OneDrive - Asia Pacific University/Documents/School Materials/Level 2/Programming for data Analytics/Assignment/Assignment Work/UNSW-NB15_uncleaned.csv"
df<-read.csv(fileUrl)
```

3.2 Libraries:

```
#load necessary libraries
library(dplyr)
library(tidyr)
library(ggplot2) #for plots
library(pROC) #auc roc evaluation
library(randomForest) #random forest
library(caret) #confusion matrix # load every time
```

dplyr and tidyr packages were used for data manipulation and cleaning, ggplot2 was used for plots. pROC and caret were used for model evaluation and randonForest was used for a randomForest classifier.

4.0 Data Cleaning

4.1 Initial Inspection, Observations and Cleaning methodologies:

Before cleaning was conducted, the class function was used to see the data type of each separate variable.

```
> class(df$service)
[1] "character"
> class(df$dload)
[1] "character"
> class(df$label)
[1] "character"
>
```

As shown above, all the variables are represented as characters which will make our analysis almost impossible as it is difficult to visualize, explore and analyse data in this format.

Service variable:

```
Console  Terminal ×  Background Jobs ×
R 4.5.2 · ~/
> table(df$service)

      -     dhcp      dns      ftp ftp-data     http      irc     pop3   radius
  92221       92    46311     3346     3931    18362       24     1085       12
   smtp     snmp      ssh      ssl
   4953       78     1277       54
```

As shown in the figure above, many services are "-" meaning they are unknown. Hence they are to be combined with any other missing values to denote they are unknown. One assumption is that all unknown types of services may show a larger frequency towards cyber-attacks. The combination and conversion are shown below:

```
 # Convert to character for manipulation
df$service <- as.character(df$service)

# Combine missing/unknown categories
df$service[df$service == "-" | is.na(df$service)] <- "unknown"
```

As seen from above service types with below 1000 such as irc, smp, ssl , radius and dhcp will be harder to analyse. It is better for them to be put under one category called "other". The more frequent ones would be easier to test and analyse; the data will be easier to visualize with the less categories we have.

```
#group rare services (<1000) into 'other'
service_count <- table(df$service)
rare_services <- names(service_count[service_count < 1000 & names(service_count) != "unknown"])
df$service[df$service %in% rare_services] <- "other"
```

The variable is then converted to a factor so that we're able to categorize the data properly:

```
# Convert back to factor
df$service <- as.factor(df$service)
class(df$service)
```

```
> df$service <- as.factor(df$service)
> class(df$service)
[1] "factor"
```

Dload variable:

The dload variable represents the download rate to destination in bits per second. The Low-rate Denial of Service LDos attacks is where attackers manipulate or completely conceal the download rate,. ("The Detection Method of Low-Rate DoS Attack Based on Multi-Feature Fusion," 2020). dload variable contained 37,917 missing values.

Rather than treating them as noise, they may indicate several types of disguised cyber-attacks. Therefore, a dload_missing_flag was derived to retain this potential anomaly.

Median imputation was used as the data is not as highly skewed, it would inflate the imputed values, distort analysis and is a more cautious approach towards outliers.

The code snippet below shows the approach mentioned as well as the median imputation done on the dload variable.

```
summary (df$dload)
   Min.  1st Qu.   Median     Mean  3rd Qu.     Max.     NA's
      0        0     1458   666357    26948 22422730    37917
```

```
#Imputing the dload with the median whilst maintaning the NA variables as new variable
df$dload<- as.numeric(df$dload)

df$dload_missing_flag <- ifelse(is.na(df$dload), 1, 0)
df$dload[is.na(df$dload)] <- median(df$dload, na.rm = TRUE)
```

Another Assumption made was that dload values of 0 might be an indicator of cyber-attacks, as shown below there are 66,048 zero values in the data set which is the most any one value holds.

```
> table(df$dload)

        0  4.297421  4.479979  4.480072  7.555602  7.763633  7.764773
    66048         1         1         1         1         1         1
```

These might also be considered as cyber-attacks. To cater for this a dload_zero_flag was derived to investigate this potential anomaly.

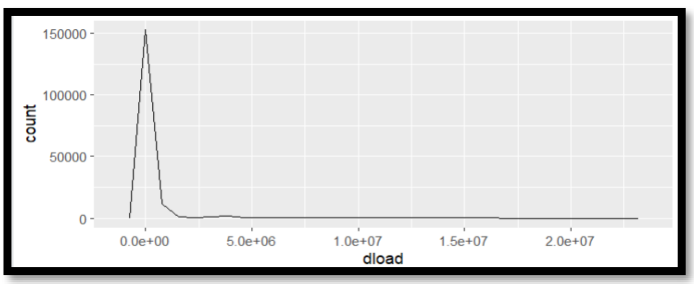
```
#flag the 0 values
table(df$dload)
df$dload_zero_flag <- ifelse(df$dload == 0, 1, 0)
```

Another way we can back our assumption is by viewing the number of 0 values within the dload variable. The summary below shows that roughly 37.7% of the values have a 0 value.

```
> df$dload_zero_flag <- ifelse(df$dload == 0, 1, 0)
> summary(df$dload_zero_flag)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
 0.0000  0.0000  0.0000  0.3767  1.0000  1.0000 
```

The plot below is a frequency polygon, it shows a high skew in the variation of the dload variable making it hard to visualize, explore, analyse, and interpret.

```
#visualizing dload

ggplot(df, aes(x=dload)) +
  geom_freqpoly()
```



This is also shown by the summary the minimum being 0 and the maximum is 22,422,730. This extreme range makes it hard to interpret as the variation high.

```
> summary(df$dload)
    Min.  1st Qu.   Median     Mean  3rd Qu. 
       0        0     1458   522575     7607 
    Max. 
22422730 
```

To combat this, we will transform the data by performing the logp, log(x+1) transformation.. The transformed variable will be represented by the variable dload_log.

```
#log-transform the dload to avoid higher skew in visualization and makes it

df$dload_log <- log1p(df$dload)   # log(1 + x), safer for zeros
```

This normalizes the distribution as there is less skew, and the values don't vary as highly. This enables better visualization and modelling.

<u>Label Variable:</u>

The label variable is categorical but is binary. "1" represents an attack, "0" represents benign traffic. As shown below there are other variants of unique data. Ideally we are to have three categories. 0, 1 and NA. The values are standardized.

```
# Standardize normal label
df$label[df$label %in% c("0", "0_", "0?")] <- "0"

# Standardize attacks
df$label[df$label %in% c("1", "1_", "1?")] <- "1"

# Set missing/invalid to NA
df$label[df$label %in% c("NA?", "NA_")] <- NA
```

The variable is then set to factor as it is categorical. There are levels applied to it so that 1 and 0 represent attacks and normal traffic respectively.

```
# Convert to factor with meaningful labels
df$label <- factor(df$label, levels = c("0", "1"), labels = c("normal", "attack"))
```

All rows NA are now eliminated leaving us with only 0 or 1, to avoid any error in our analysis it was decided that NA values are dropped as we conclude as to whether they are attacks or not.

```
# remove rows with NA label
df <- df[!is.na(df$label), ]
```

5.0 Data Validation

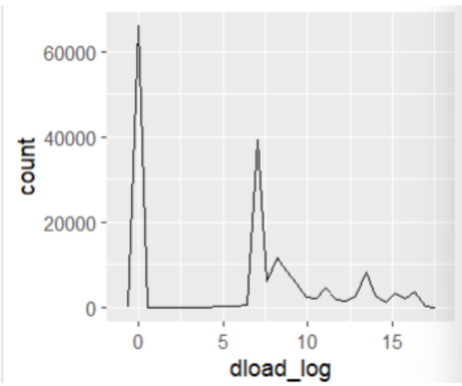
The following was done to validate the cleaned data and confirm it is ready for analysis and interpretation.

As shown below the service variable has been cleaned as required showing 9 different categories. The summary shows each category and how frequently they occur the modal category is unknown.

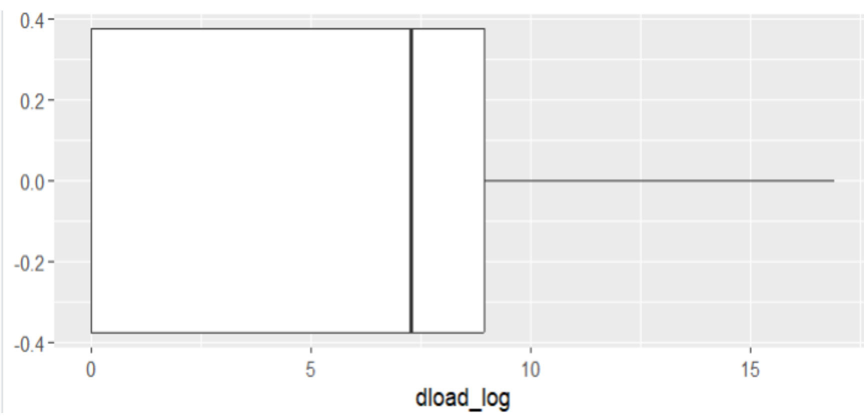
```
> summary (df$service)
     dns      ftp ftp-data     http    other 
   46311     3346     3931    18362      260 
    pop3     smtp      ssh  unknown 
    1085     4953     1277    95816 
```

The newly derived dload_log variable has a reduction in skew and has a lower variation than the original dload. The median imputation makes sure that outliers are reduced. The new values found are shown below together with a frequency plot that is inclusive of all the varying data with a smaller range.

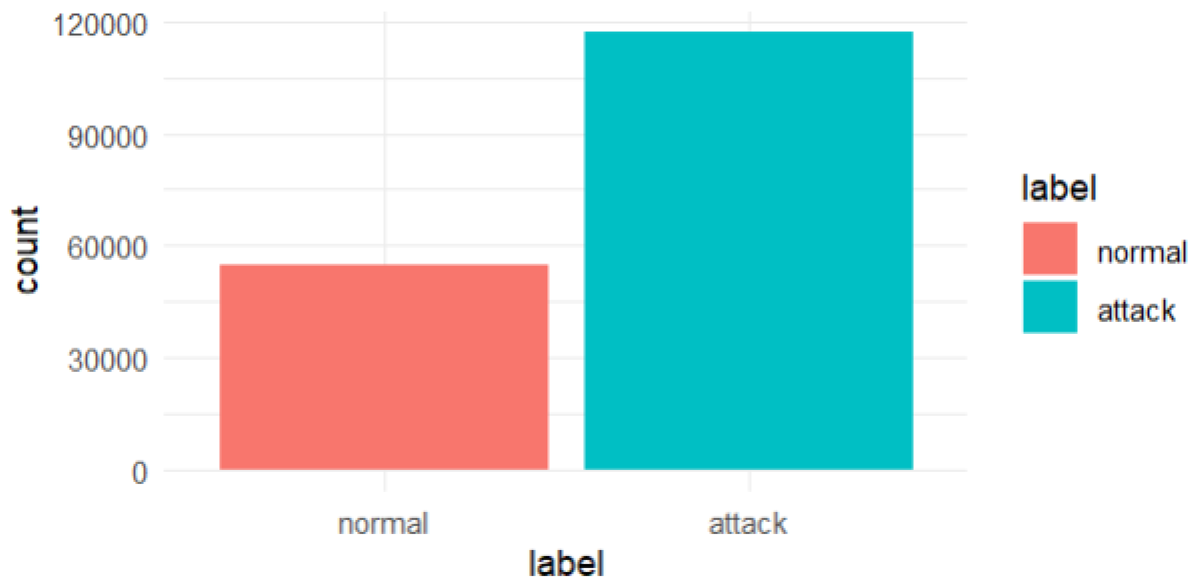
```
> summary(df$dload_log)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  0.000   0.000   7.285   5.977   8.937  16.926 
```



The box plot below properly shows the 1st quartile, the 3rd quartile, the quartile range as well as the median, maximum and median values. As observed there are no existing outliers. This also ensures that any modelling will be more accurate as it reduces the likelihood of a high variance, low bias.



The label variable was eradicated from NA's and strange variants, leaving only 0 and 1 to represent benign traffic and attacks respectively, levels "normal" and "attack" were assigned to for easier representation as shown by the bar plot and summary below.




```
> summary(df$label)
normal attack
 54858 116960
```

6.0 Data Analysis

6.1 Exploratory Data Analysis:

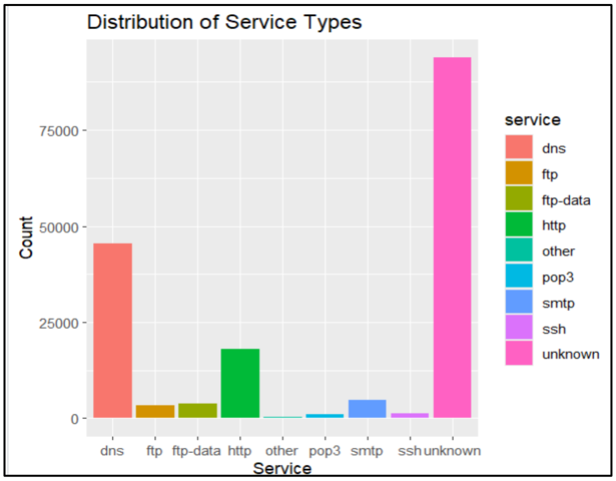
The following exploratory data analysis was carried out to perform the following functions.

1.) Single variable analysis to understand the distribution of each variable.
2.) Bivariate and multivariate analysis to understand the relationships between different variables and how they associate with cyber-attacks.

Single Variable:

Service distribution:

```
ggplot(df,aes(x=service, fill=service))+
  geom_bar()+
  labs(title = "Distribution of Service Types", x="Service",y="Count")

table(df$service)
```




```
    dns       ftp ftp-data      http     other      pop3
  45403      3277     3857     17982       256      1055
   smtp       ssh  unknown
   4858      1253    93877
```

In the bar plot and the tables in the figures above. The unknown services dominate the plot as there are around 93877, with dns service being the second most frequent. Altogether we can see that there are 9 different service categories.

The table reflects the exact number of service categories that exist in the data set.

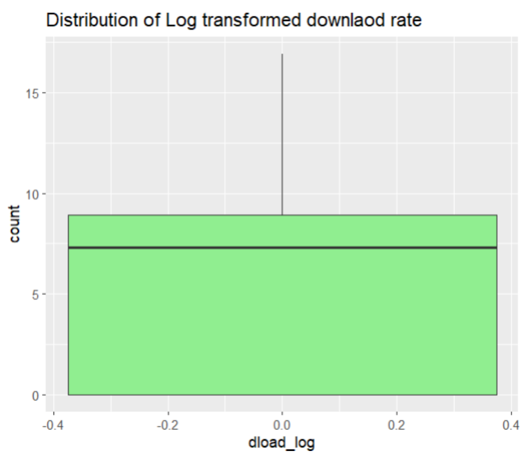
<u>dload_log variable:</u>

dload_log is the log transformed version of dload as it is easier to visualize due to the higher distribution of values. The dload log variable is a continuous variable.

```
> summary(df$dload_log)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  0.000   0.000   7.285   5.976   8.937  16.926 
```

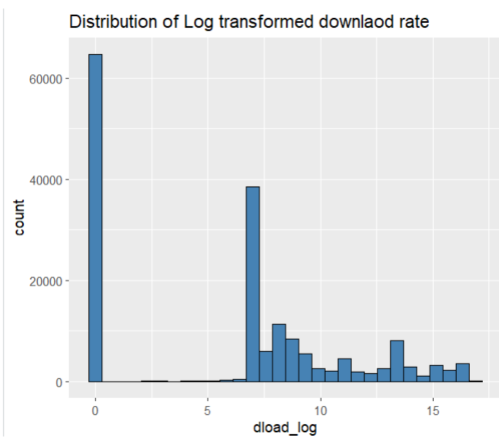
The summary above shows us that the 1st Quartile of the data is at 0, the 0 values were not neglected during the data cleaning process as they might be a significant to the presence of cyber-attacks. The minimum and maximum values are 0 and 16.926. The mean is 5.976 and the 3rd quartile is 8.937. The median value is 7.285

```
ggplot(df, aes(y=dload_log))+
    geom_boxplot(fill="lightgreen")+
    labs(title="Distribution of Log transformed downlaod rate",x="dload_log", y="count")
```



The boxplot above shows us that there are no existing outliers in the dload_log variable, reducing the potential of a high skew and distorting model performance which could cause overfitting in a model.

```
ggplot(df, aes(x=dload_log))+
  geom_histogram(bins=30,fill="steelblue",color="black")+
  labs(title="Distribution of Log transformed downlaod rate",x="dload_log", y="count")
```



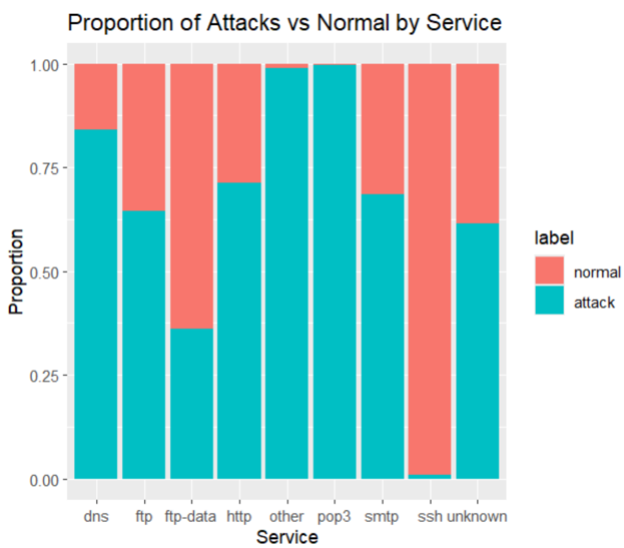
The histogram above shoes a zero inflated distribution. As the first quartile and minimum are both 0. This indicates that majority of the observations, (roughly around 6,600) have no download activity. The skew observed leans more towards the right. The median value being around 7.3, with this observation it is easy to note that the mean is less than the median value. The non-zero values extend toward a maximum of 16.0 which forms a longer tail on the right. The pattern above suggests that most network traffic exhibit very low activity.

<u>Bivariate and Multivariate analysis:</u>

```
> table(df$service, df$label)

          normal attack
  dns       7211  38192
  ftp       1166   2111
  ftp-data  2468   1389
  http      5144  12838
  other        3    253
  pop3         3   1052
  smtp      1526   3332
  ssh       1242     11
  unknown  36095  57782
```

The cross tabulation of service and label shows that a majority of the cyber-attacks come from unknown, dns and http service types with unknown being the majority at 57,782. dns and http have 38,192 and 12,838 respectively.

```
# Proportional bar plot
ggplot(df, aes(x=service, fill=label)) +
  geom_bar(position="fill") +
  labs(title="Proportion of Attacks vs Normal by Service", x="Service", y="Proportion")
```

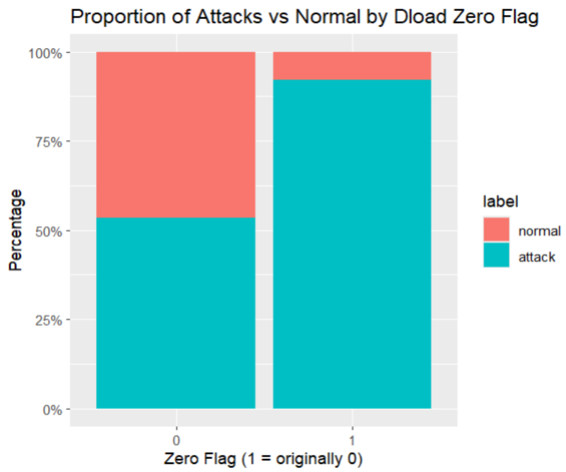


The bar plot shows the proportion of normal and attacks to the type of service. Y-axis ranges from 0.00 and 1.00 with 0.00 representing 0% and 1.00 representing 100% for each service category. We can see that approximately 85% of the dns service in the dataset is associated with cyber-attacks. Nearly 63% of the unknown service are attacks. For the services classed as other and pop3, show high attack rates estimated 96-98% of these types of services are attacks.

The ftp-data service shoes that nearly 35% of them are attacks and about 65% of the ftp service show an inclination towards attacks. The smtp service has roughly 68% being attacks. Overall, the proportional bar plot indicates that the majority of service types in the dataset have a strong tendency towards attack activity.

<u>dload and it's variants.</u>

<u>dload_zero_flag and label</u>

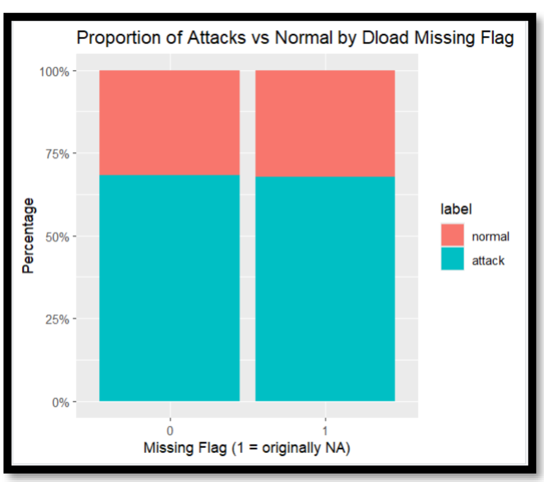


The above shows a barplot of dload_zero_flag and label. 0 represents the dlaod values that did not have a zero value, whilst 1 represents those that had a zero value. This was done to analyze the potential of the Low-rate Denial of Service attacks.

As shown above, the relationship between the two variables suggest that approximately 89% of the values that were flagged as 0 were associated with cyber-attacks, whilst approximately only 54% of the values that weren't flagged as 0 were associated with cyber-attacks.

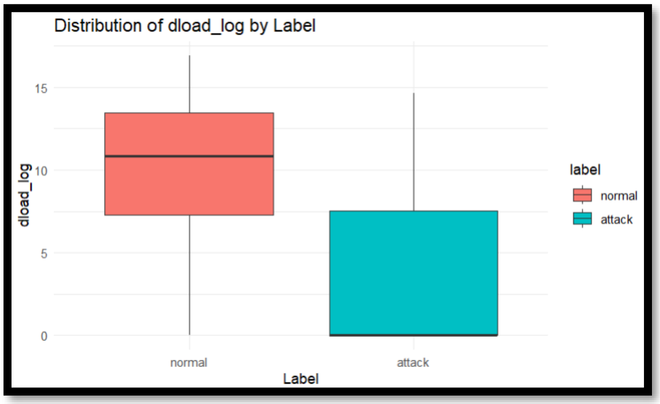
With this information we can work on the assumption that majority of the zero flagged values contribute to cyber-attacks, primarily the low-rate denial of service.

<u>dload_missing flag and label:</u>



From the above we can see the relationship between dload_missing_flag and label. About 69% of the records with no missing values are associated with cyber-attacks. Approximately 69% of the values that were flagged as missing are associated with cyber-attacks. This indicates that missing dload values do not behave differently from non-missing ones and still occur in attack traffic.

dload_log and label.



As seen above, normal sessions have a median of approximately 11, a lower bound of approximately 7.5, and an upper bound of 13.5 and whiskers that run from 0 - 16.9. Attack sessions show a median dload_log of 0, where Q1 is approximately 0 and Q3 is approximately 7.5, and whiskers that run from 0 to approximately 14.8.

The cyber-attack distribution's very low median and Q1 imply that malicious sessions involve minimal download activity. While it's larger IQR and wide whisker reflect heterogeneous attack behaviors. Normal traffics higher median and narrower IQR reflect more sustained and consistent download activity.

Sessions are generally higher and more stable in benign traffic. Whereas attack sessions are often near zero but are more variable.

6.2 Statistics:

Objective: To determine whether service type and download rates (dload and other variants) significantly influence the presence of cyber-attacks.

Question: Is service type independent or dependent in relation to cyber-attack presence?

H_0: Service is independent of attack presence.

H_1: Service type is associated with attack presence.

```
> # Contingency table
> service_label_table <- table(df$service, df$label)
> # Chi-square test
> chisq.test(service_label_table)

        Pearson's Chi-squared test

data:  service_label_table
X-squared = 12352, df = 8, p-value <
2.2e-16
```

Since the p-value is far below the significance level α = 0.05, we reject the null hypothesis. This means there is a significant relation between service type and cyber-attack presence. This denotes that services are disproportionately associated with attacks, this confirms service type is an indicator towards the presence of cyber-attacks.

Question: Is the download rate independent or dependent in relation to cyber-attack presence?

H_0: download rate is not associated with attack presence.

H_1: download rate is associated with attack presence.

```
        Wilcoxon rank sum test with continuity
        correction

data:  dload_log by label
W = 5218100981, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal
to 0
```

The p-value is far below 0.05, so the null hypothesis that download rates dload_log are the same for attack and normal sessions. From the EDA, normal traffic has a higher median and more consistent variation, while attack sessions cluster near zero but are more variable. dload_log is a strong feature towards the presence of cyber-attacks.

6.3 Logistic Regression:

Do service type and download related features (dload_log, dload_zero_flag, dload_missing_flag) significantly influence the likely presence of cyber-attacks?

```
# split 70% train, 30% test
index <- sample(1:nrow(df), size = 0.7 * nrow(df))
train <- df[index, ]
test  <- df[-index, ]
```

The above shows the split train and testing, the dataset was randomly split into a training set (70%) and a testing set (30%) to validate model performance.

A generalized logistic regression model (glm) was fitted on to the training set with the dependent variable being label as shown below.

```
> #generalized logistic regression model encompassing all variables
> model <- glm(label ~ service + dload_log + dload_zero_flag + dload_missing_flag,
+              data = train,
+              family = binomial)
> summary(model)

Call:
glm(formula = label ~ service + dload_log + dload_zero_flag + 
    dload_missing_flag, family = binomial, data = train)

Coefficients:
                    Estimate Std. Error  z value Pr(>|z|)    
(Intercept)         7.631331   0.071080  107.362  < 2e-16 ***
serviceftp         -0.876950   0.052825  -16.601  < 2e-16 ***
serviceftp-data    -1.311534   0.051481  -25.476  < 2e-16 ***
servicehttp         0.146853   0.032814    4.475 7.63e-06 ***
serviceother        3.146946   0.723761    4.348 1.37e-05 ***
servicepop3         6.145740   0.577159   10.648  < 2e-16 ***
servicesmtp        -0.131182   0.048865   -2.685  0.00726 ** 
servicessh         -5.249243   0.352418  -14.895  < 2e-16 ***
serviceunknown     -1.351805   0.024891  -54.308  < 2e-16 ***
dload_log          -0.681348   0.006257 -108.899  < 2e-16 ***
dload_zero_flag    -4.267253   0.064539  -66.119  < 2e-16 ***
dload_missing_flag -1.043650   0.024341  -42.877  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 150694  on 120271  degrees of freedom
Residual deviance:  97165  on 120260  degrees of freedom
AIC: 97189

Number of Fisher Scoring iterations: 7
```

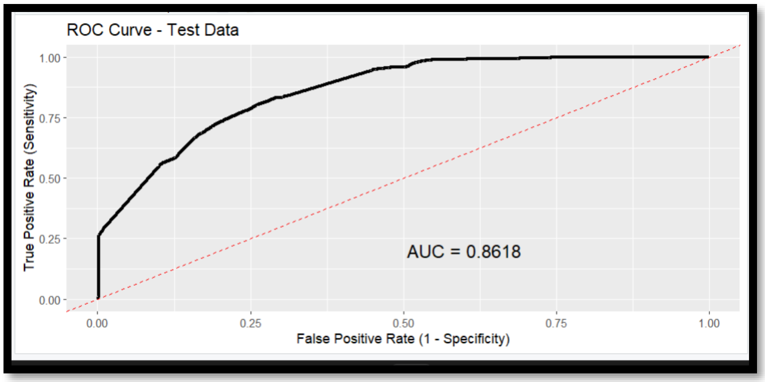
All predictor variables are statistically significant ($p<0.01$). Zero and missing download flags were strong indicators towards the presence of cyber-attacks, while higher log-transformed download rates decreased the likelihood. Service types such as other and pop3 were strongly associated with attacks.

Predictions were generated for the test data set and evaluated using the ROC and AUC metric.

```
#predicting
prob_test <- predict(model, newdata = test, type = "response")
```

```
> prob_test <- predict(model, newdata = test, type = "response")
> roc_test <- roc(test$label, prob_test)
Setting levels: control = normal, case = attack
Setting direction: controls < cases
> auc(roc_test)
Area under the curve: 0.8618
```

The model achieved an AUC (Area Under the Curve) of 0.8618 on the test set demonstrating excellent discriminative capacities, towards the differentiation of benign and attack traffic. This means that there is an 86% probability that the model will assign a higher likelihood of attack to a randomly chosen attack.



The ROC curve above visually illustrates the model's performance in distinguishing cyber-attacks from benign traffic. The dashed red line shows a random classifier (AUC=0.5). The ROC curve is high above the random classifier as the curve runs steeply. Both the curve and the AUC indicate excellent discrimination (*What Is a Good ROC Curve Score?*, 2022).

The logistic regression model shows us that all the variables are associated towards the presence of cyberattacks with some service categories showing more association than others.

<u>6.4 Random Forest (Extra Feature):</u>

```
#Random Forest
rf_model <- randomForest(label ~ service + dload_log + dload_zero_flag + dload_missing_flag,
                         data = train,
                         ntree = 1000,          # number of trees
                         mtry = 2,             # number of variables considered at each split
                         importance = TRUE)    # allows us to see feature importance
print(rf_model)
```

```
Call:
 randomForest(formula = label ~ service + dload_log + dload_zero_flag +      dload_missing_flag,
data = train, ntree = 1000, mtry = 2,      importance = TRUE)
               Type of random forest: classification
                     Number of trees: 1000
No. of variables tried at each split: 2

        OOB estimate of  error rate: 16.25%
```

Random Forest is a linear ensemble model that captures complex relationships beyond the other techniques used as variable importance plots highlight which features influence the presence of cyber-attacks.

A Random Forest classifier with 1,000 trees and *mtry =2 (based on* $\sqrt{p}$ *= 2 predictors per split)* was trained to further assess variable influence. The OOB estimate gave an error rate of 16.25% meaning that 83.75% of the training data is correctly classified.

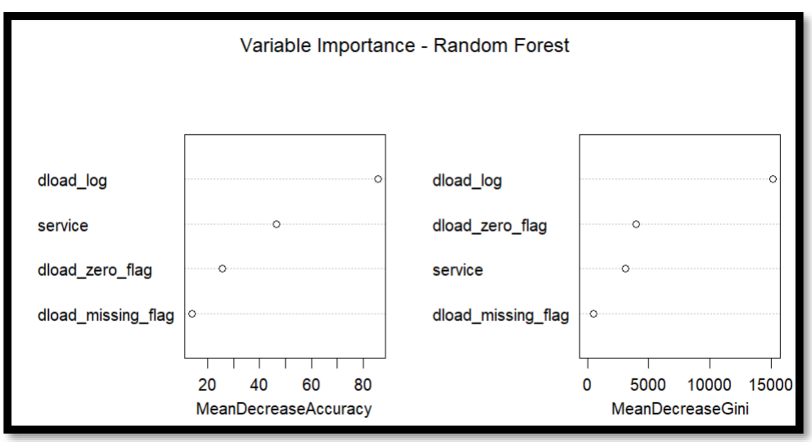
```
#predictions on test set
rf_pred <- predict(rf_model, newdata = test)
```

```
> conf_mat <- confusionMatrix(rf_pred, test$label)
> print(conf_mat)
Confusion Matrix and Statistics

          Reference
Prediction normal attack
    normal   8899    644
    attack   7535  34468

               Accuracy : 0.8413
                 95% CI : (0.8381, 0.8445)
    No Information Rate : 0.6812
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.5888

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.5415
            Specificity : 0.9817
         Pos Pred Value : 0.9325
         Neg Pred Value : 0.8206
             Prevalence : 0.3188
         Detection Rate : 0.1726
   Detection Prevalence : 0.1851
      Balanced Accuracy : 0.7616

       'Positive' Class : normal
```

Predictions were then made on the test data, the results were shown above as the evaluation of the model was given by the confusion matrix. The model showed a strong performance achieving an accuracy of 84.13% on the test data. The model demonstrated great ability to detect cyber-attacks with a specificity value of 98.17%, it correctly classified almost all attack instances. However, the sensitivity was lower with 54.15% indicating that more than of benign traffic was misclassified.

From a cyber-security perspective this trade-off is more of an upside. Minimizing false negatives is more critical than minimizing false positives. A balanced accuracy of 76.16% and a kappa of 0.589 suggests that the model performs reliably despite dataset imbalance.

```
#variable importance
varImpPlot(rf_model, main = "Variable Importance - Random Forest")
```



The plot above represents the variable importance in the random forest model. The most influential predictor by far is the dload_log as shown by its high mean decrease in accuracy, and it's mean decrease gini scores.

In contrast, variables such as service and dload_zero_flag add moderate but small value to the model, which dload_missing_flag has the lowest importance and contributes minimally to prediction.

From the analysis run with the use of Random Forest, we can conclude that cyber-attack detection is highly influenced by behavioural network features. The dload_log variable is the most influential when it comes to the correlation to cyber-attacks.

7.0 Conclusion

This project provided a data analysis of the UNSW-NB15 dataset to investigate whether the service type and download behaviour influence the presence of cyber-attacks. Through data cleaning, exploratory data analysis, using statistical tests and machine-learning clear relationships between network behaviour and attack activity were identified.

Both statistical test and predictive models confirmed that service categories and download related features are related significantly with cyber-attacks. In particular the dload_log variable was the most influential predictor. The logistic regression model achieved strong discriminatory performance with an area under the curve of 0.8618, indicating that behavioural network features can effectively support intrusion detection.

Overall, the results show that network behaviour characteristics can be successfully employed to categorize and identify cyber-attacks within modern intrusion detection systems.

7.1 Findings

From the investigation and the analysis carried out. Cyber-attack detection is strongly influenced by behavioural network features. dload_log is the most important predictor. Attacks overall exhibit higher variability and abnormal patterns as shown by statistical tests. Skewed distributions and class imbalance show the need for careful preprocessing and feature engineering to improve intrusion detection performance.

7.2 Recommendation

Intrusion detection systems should incorporate behavioural network indicators, download rate patterns and service usage in particular.

Addressing class imbalance using techniques as SMOTE, feature scaling and hyperparameter tuning would yield better unbiased results. Furthermore, the use of more advanced algorithms such as gradient boost or deep learning would increase accuracy with finer differences. Finally, more appropriate real-time models would be those that are frequently evaluated and retrained to match the dynamic setting of a network and trends in attacks.

7.3 Limitations

This study analysed a limited number of features which restricted the overall predictive capability of the models

The random forest model was implemented without hyperparameter tuning and with a small amount of variables which decreased the potential for accurate predictive power. Finally, the dataset was also stale and imbalanced, meaning the generalizability of results was not as effective in comparison to real-time networks.

7.4 Future Directions

Future projects could focus on the incorporation of additional network features such as protocol-specific metrics, to improve sensitivity and detect more subtle attacks.

Exploring advanced machine learning techniques such as gradient boost deep learning models, may refine composite features and may enhance predictive detection performance and capture complex attack patterns.

Addressing class imbalance with techniques like oversampling or adaptive thresholds, validating models on real-time datasets and using explainable AI methods could improve generalization and interpretability. Ultimately more robust techniques could support better and more reliable intrusion detection for cyber security analysts.

8.0 References

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